Spatio-temporal Analysis of Influenza Epidemics in Japan

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Abstract

An epidemic of influenza is a complicated phenomenon influenced by many social and biological factors such as geography, climate, populations, transport network, biological statuses of human and virus, and so on. To investigate these influences, this study analyzed epidemiological data of influenza incidences in Japan from Apr 1999 to Dec 2012 by use of wavelet. We calculated power spectra, and derived a contour plot of phase differences (Fig.1). This plot exhibits band structures in both space and time. The band structure in space shows the following features: (1) the epidemics synchronized strongly in 7 prefectures around Tokyo, and the epidemics in 7-8 prefectures around Osaka also synchronized, but with another phase; (2) except these urban epidemics, the epidemics propagated from western to eastern prefectures, and finally to northern prefectures (broken guideline); (3) Miyagi prefecture (No.4) has an epidemic phase different from the other northern prefectures; (4) although Yamanashi (No.19) is a prefecture neighboring Tokyo, there is a discrepancy between their epidemic phases. In the meeting, we will present our interpretation and other interesting features obtained from our analysis.

Keyword: Influenza, Wavelet, Epidemiology, Data analysis

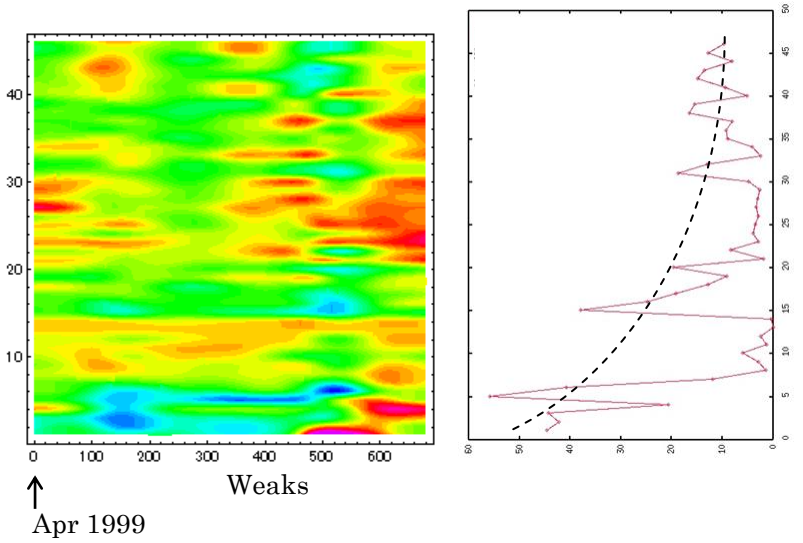


Fig.1 Contour plot of epidemic-phase differences that are differences from the phase in Tokyo (left-hand panel), and the time-averaged phase differences for the prefectures (right-hand panel). The numbers in the vertical axis indicate the prefectures (Tokyo=13, Osaka=26); the numbers are arranged roughly from northern, eastern to western prefectures.